

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

April 2013

EC 203 Solid State Electronics

Date: 18.04.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define and describe insulator, semiconductor and metal with energy level diagram. [03]
(b) Write brief note on Donor and Acceptor impurities. [04]
- Q - 2 (a) Explain forward bias and reverse bias operation for p-n junction diode. [04]
(b) What is mobility? Derive equation for current density J and conductivity σ . [05]
(c) Prove that Fermi level lies in center of forbidden energy band in intrinsic semiconductor. [05]

OR

- Q - 2 (a) What is breakdown region in case of diode? Also Explain reverse saturation current. [04]
(b) Derive the equation for space-charge capacitance C_T . [05]
(c) What is mass action law? In a p-type silicon sample the hole concentration is $2.25 \times 10^{15}/\text{cm}^3$. If the intrinsic carrier concentration is $1.5 \times 10^{10}/\text{cm}^3$, find the electron concentration. [05]
- Q - 3 (a) Explain in brief – Zener Diode. [04]
(b) Explain CB Configuration for BJT with concept of Early Effect. [05]
(c) Determine the following for the network drawn in diagram-1 with $I_{DSS}=10 \text{ mA}$ and $V_P=-8 \text{ V}$ [05]
(1) I_{DQ}
(2) V_{DS}
(3) V_G

OR

- Q - 3 (a) Explain in brief – Semiconductor Photodiode. [04]
(b) Explain CE Configuration for BJT with all three Q-point operating region. [05]
(c) The self bias configuration has operating point defined by $V_{GSQ} = -2.6 \text{ V}$ and $I_{DQ}=2.6 \text{ mA}$ with $I_{DSS}=8 \text{ mA}$ and $V_P= -6 \text{ V}$. The network is drawn in diagram-2 which has $Y_{OS}= 20 \mu\text{s}$. Find the following Parameter [05]
(1) g_m ,
(2) r_d ,
(3) Z_i

SECTION – II

- Q - 4 (a) Explain Ebers-Moll Model. [03]
(b) Explain Miller's theorem and its dual. [04]

- Q - 5 (a) Explain Hybrid model. Determine H parameter from its characteristics. [07]
(b) Explain FET construction, operation and static characteristics. [07]

OR

- Q - 5 (a) Discuss validity of hybrid Π model and parameter variation. [07]
(b) Explain enhancement type MOSFET and Compare P-channel MOSFET with N-channel MOSFET. [07]

- Q - 6 (a) Draw and explain common-source and common drain amplifiers with necessary equations. [07]
(b) Derive the Fermi Level equation for extrinsic semiconductor. [07]

OR

- Q - 6 (a) Draw and write equations for Fixed Bias, Self Bias and voltage divider bias. [07]
(b) Derive the electron and hole concentration equation for intrinsic semiconductor material. [07]


